



Name _____ HR _____

7

7th Grade Science Unit 7: Engineering STAR Cards!

QUESTION	ANSWER																																		
7a) ☆☆☆ When thinking of their design solution, engineers need to think about the <u>criteria and constraints</u> of their design. What are criteria and constraints?	7a) ● <u>Criteria</u> = requirements of the design ● <u>Constraints</u> = limitations of the design																																		
7b) ☆☆☆ Why do engineers test and evaluate prototypes?	7b) Engineers test and evaluate prototypes to make the design solution better.																																		
7c) ☆☆☆ What is the purpose of using a <u>decision matrix</u> like the one below? <table border="1"><thead><tr><th rowspan="2">Type of Block</th><th colspan="4">Criteria</th><th rowspan="2">Total Points</th></tr><tr><th>Low Cost</th><th>Easy to Build With</th><th>Durable</th><th>Able to Stick Together</th></tr></thead><tbody><tr><td>Number cubes </td><td>3</td><td>2</td><td>4</td><td>1</td><td>10</td></tr><tr><td>Sugar cubes </td><td>4</td><td>1</td><td>1</td><td>1</td><td>7</td></tr><tr><td>Interlocking cubes </td><td>2</td><td>3</td><td>4</td><td>4</td><td>13</td></tr><tr><td>Modeling clay </td><td>2</td><td>3</td><td>2</td><td>3</td><td>10</td></tr></tbody></table>	Type of Block	Criteria				Total Points	Low Cost	Easy to Build With	Durable	Able to Stick Together	Number cubes 	3	2	4	1	10	Sugar cubes 	4	1	1	1	7	Interlocking cubes 	2	3	4	4	13	Modeling clay 	2	3	2	3	10	7c) A decision matrix helps engineers to evaluate and select the best design based on given criteria
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